



3D-BreastNet: A Self-supervised Deep Learning Network for Reconstruction of 3D Breast Surface from 2D Thermal Images

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Abstract. Infrared thermography is gaining attention as an affordable, portable, privacy-sensitive, and radiation-free breast imaging modality. However, thermal images alone provide only two-dimensional (2D) information necessitates multiple images to visualize the complete 180° breast region, posing challenges in visually correlating these images. To address this issue, we propose 3D-BreastNet, a deep-learning architecture that reconstructs 3D thermal images from 2D thermal images. The proposed method employs a self-supervised learning strategy to learn the 3D silhouette and overlay observed temperatures onto the predicted 3D structure. To evaluate the performance of 3D-BreastNet, we compared the 2D projections of the predicted 3D breast silhouette at standard view angles with their corresponding input 2D silhouettes. The accuracy, dice index, jaccard index, and hausdorff distance were found to be 0.97, 0.97, 0.94, and 3.86, respectively, on an independent test set comprising 1000 thermal images from 200 participants. The results obtained with 3D-BreastNet demonstrate promising outcomes, highlighting the feasibility of reconstructing 3D temperature information from 2D thermal images alone. This research opens new possibilities for enhanced breast imaging and diagnosis using deep learning techniques.

Keywords: 3D Thermal Image · 3D Reconstruction · 3D Registration · Deep Learning · Self-supervised Learning

1 Introduction

Infrared thermography has gained recognition as a valuable tool for medical imaging, offering benefits such as non-invasiveness, affordability, and radiation-free imaging. It has been used for various applications, including breast cancer diagnosis, treatment monitoring, and pain management [1]. In recent times, infrared thermography has emerged as a potential screening tool for COVID-19 in public places like airports and hospitals [2]. Advancements in uncooled microbolometer Focal Plane Arrays (FPAs) have played a pivotal role in re-establishing the use of infrared thermography in medical applications since the 1990s. Modern thermal imaging cameras can detect temperature variations as low as 20 mK with high spatial resolution and are available at affordable costs [3].

Breast cancer is a major cause of cancer-related deaths among women globally [4]. Early detection is crucial for improving survival rates, but low- and middle-income countries face challenges in implementing conventional imaging modalities like mammography, MRI, and ultrasound due to cost and infrastructure constraints [5–8]. The increased metabolic activity in cancerous tissues leads to elevated heat generation, which can be detected through thermal imaging [9–13]. By leveraging the advantages of affordability and portability, thermal imaging has the potential to serve as an alternative modality for breast cancer detection, particularly in resource-constrained settings [14]. Breast thermography has been vastly explored in the literature for breast cancer diagnosis [15, 16], treatment monitoring [12, 17], prognosis [18], breast augmentation [19], and 3D bioheat modeling of cancerous regions.

While thermal imaging provides valuable two-dimensional (2D) temperature distribution information, it lacks the spatial characteristics of three-dimensional (3D) thermal patterns. To overcome this limitation, multiple 2D thermal views are captured to cover the entire breast region. However, visually correlating these images poses challenges, requiring mental mapping of large amounts of pixel information. The availability of 3D spatial information is crucial for accurate tumor depth estimation, 3D bio-heat modeling, tumor orientation analysis, structural property assessment, and breast augmentation.

Existing research on the automated generation of 3D thermal images from 2D thermal images is limited. Some approaches utilize additional hardware [20–24], such as 3D scanners, which increase complexity and imaging concurrence requirements. Other methods rely on image processing techniques [25] that approximate 3D shapes from body contours alone, potentially resulting in imperfect 3D reconstructions due to the non-rigid and amorphous nature of the breast. In recent years, deep learning [26–28] has shown promising results in 3D object and scene reconstruction from 2D images without explicit 3D ground truth. These approaches leverage information about captured view angles and spatial distances between pixels to infer the missing dimension. Building upon these advances, we propose a deep learning framework called 3D-BreastNet to reconstruct 3D thermal images of the breast surface from 2D thermal images captured at different view angles, without relying on 3D ground truth or precise view angle information.

Standard breast thermography protocols involve capturing multiple 2D thermal images at various view angles to cover the 180° breast region. However, challenges arise in capturing images at the precise standard view angles due to practical clinical settings, patient movement, and communication issues. With the promising results of deep learning-based 3D reconstruction in other imaging domains, it becomes feasible to generate 3D temperature information of the breast surface using deep learning solely from 2D thermal images. This paper presents the 3D-BreastNet framework as a solution to reconstruct 3D thermal images of the breast surface from 2D thermal images captured at different view angles, overcoming challenges associated with standard view angle deviations and the lack of 3D ground truth.

2 Methods

2.1 Dataset Description

We collected an anonymized dataset comprising thermal images from 2027 participants in collaboration with multiple hospitals and diagnostic centers in India. Ethical approvals were obtained from the respective institutional review boards, and participants provided signed informed consent forms. For each participant, we captured five thermal images at standard view angles: right lateral (RL) at -90° , right oblique (RO) at -45° , frontal (F) at 0° , left oblique (LO) at 45° , and left lateral (LL) at 90° (illustrated in Fig. 1(a)).

The thermal imaging protocol involved several steps. Firstly, participants were cooled in an air-conditioned room maintained at a temperature range of 22°C to 24°C to eliminate extraneous heat patterns prior to imaging. We utilized a fixed rotating chair with position markers to facilitate easy and comfortable rotation of the participants. The camera height and distance were adjusted individually for each participant, and verbal instructions were given to ensure participants turned to the standard view angles for image capture. To minimize translational and zoom variances, both the camera and chair remained fixed throughout the imaging process for each participant.

Although we followed the imaging protocol diligently, the collected thermal images deviated slightly from the standard view angles due to the inherent challenges of capturing images precisely at those angles. We employed the FLIR E75 (FLIR, Wilsonville, US) thermal camera, which offers a thermal sensitivity of 50 mK, a spatial resolution of 320×240 , and a field of view of $24^\circ \times 18^\circ$.

2.2 Proposed Methodology

Traditional supervised algorithms for 3D breast surface reconstruction typically require 3D breast thermal surfaces as ground truth, which are challenging to obtain due to the absence of publicly available datasets. To address the limitations of requiring labeled datasets with 3D ground truth, we propose a self-supervised deep learning framework named 3D-BreastNet, as illustrated in Fig. 1.

The 3D-BreastNet framework consists of five main steps: i) foreground segmentation from input thermal images to segment 2D breast silhouettes, ii) 3D breast silhouette generation from the segmented 2D breast silhouettes, iii) a 2D view generator to obtain random 2D silhouette projections from the generated 3D breast silhouette, iv) loss function to minimize the distance between the generated 2D projections and the actual silhouettes of the input 2D thermal images, and v) overlay function to map the input temperature values onto the generated 3D silhouette (3D volume).

2.2.1 Foreground Segmentation

To extract the breast region from the input thermal images, a binary segmentation process is employed using Otsu's method. This converts the raw temperature values into binary images, where '0' represents the background pixels and '1' represents the foreground pixels. This segmentation process is applied to each of the five thermal views $T_i, i \in [1, 5]$ captured at (RL, RO, F, LO, and LL), resulting in corresponding 2D breast silhouettes (I_i).

2.2.2 3D Silhouette Generator

The goal of the 3D silhouette generator is to learn a reconstruction function $G : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, that generates the 3D silhouette of the participant based on the five binary 2D silhouettes. We employ an encoder-decoder CNN architecture, as proposed in [26], to accomplish this task. The architecture consists of five up-sampling and five down-sampling stages, followed by a sigmoid activation layer to generate the 3D breast silhouette. The five 2D silhouettes obtained from the foreground segmentation are stacked along the z-axis (channels) to form the input to the encoder.

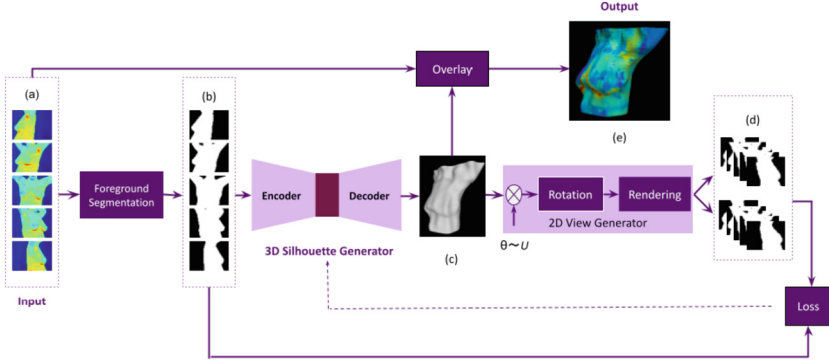


Fig. 1. Illustration of the 3D-BreastNet framework for the reconstruction of a 3D surface from 2D thermal images. a) Input thermal images, b) Binary 2D silhouettes, c) Generated 3D silhouette, d) Random 2D projections, e) 3D Surface temperature obtained by overlay.

2.2.3 2D View Generator

The 2D view generator module aims to generate random 2D projections from the 3D silhouette and utilize these projections to assess the accuracy of the generated volume. Since this learning task does not involve explicit 3D labeled data, the training of 3D-BreastNet becomes a self-supervised learning process. To obtain random 2D projections, a transformation is employed to project the 3D silhouette onto 2D planes at different angles as shown in Eq. 1.

$$S_i = T_r(V, \theta_i) \quad (1)$$

S_i represents the 2D projection silhouette, V represents the generated 3D silhouette, $\theta_i \in [-90^\circ, 90^\circ]$ represents random view angle, and T_r is a transformation that projects V to a 2D plane along angle θ_i . This transformation can be approximated as a sequential rotation of the volume followed by projection onto the frontal plane. These two steps of rotation and rendering can be mathematically formulated as below.

- Rotation:

To perform rotation, a three-dimensional Euclidean space with the z-axis representing the depth, the y-axis representing the vertical axis, and the x-axis representing the

horizontal axis is considered. In this assumed Euclidean space, the participant would be rotating about the y-axis (vertical axis) and therefore, the rotation transformation matrix is written as:

$$R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \quad (2)$$

Using the above rotation transformation matrix $R_y(\theta)$, the points on the generated 3D silhouette are rotated by an angle θ , such that the frontal view in the rotated volume, now emulates the view corresponding to view angle θ .

- **Rendering:**

To obtain a 2D projection image from the rotated volume V_θ , we map the 1D voxel vectors along the z-axis in V_θ to a 2D pixel value using a transformation $\rho : R^3 \rightarrow R^2$, as described in Eq. 3. Since the generation of a 2D silhouette (binary image) of the participant is of particular interest, a type of volume rendering known as visual hull [26] that converts scalar density voxels into binary opacity pixels is used for the rendering transformation ρ .

$$\rho(v) = 1 - \exp\left(\sum_{k=0}^{k=|v|} -v_k\right) \quad (3)$$

In the above equation, v represents 1D voxel vector consisting of voxels along the z-axis and $|v|$ represents the cardinality of v , which also correspond to the depth of volume V . A pixel value of ‘0’ for $\rho(v)$ represents empty space and a pixel value of ‘1’ represents fully occupied. This rendering transformation is applied on all the voxel vectors in the volume to obtain the 2D silhouette projection image.

2.2.4 Loss

Obtaining 3D ground truth volumes for breast thermography is challenging. This limitation prohibits the use of supervised learning algorithms that rely on comparing predicted volumes with actual 3D ground truth volumes. Additionally, breast thermography imaging lacks standardization, leading to deviations from the standard view angles (0° , $\pm 45^\circ$, $\pm 90^\circ$). To address these issues, 3D-BreastNet trains the 3D silhouette generator (G) to ensure that the 2D projections of the generated 3D silhouette at random view angles closely match the input 2D silhouette images. The training process involves minimizing the Dice loss, as represented in Eq. (7), which compares the 2D rendered projection (S) of the generated 3D volume with the input 2D silhouette (I). This approach eliminates the need for precise view information and mitigates view errors that occur during breast thermography imaging capture.

To obtain the random projections at random view angles θ_{ij} , we first randomly select n views for each of the standard view angles through uniform sampling (U) on the upper (u) and lower (l) limits as mentioned in below Eq. (4) and (5):

$$\theta_{ij} = [U(\theta_{li}, \theta_{ui})]_j \quad (4)$$

$$(\theta_{l_i}, \theta_{u_i}) = \begin{cases} (-90^\circ, -67.5^\circ), & i = 1 \\ (-67.5^\circ, -22.5^\circ), & i = 2 \\ (-22.5^\circ, 22.5^\circ), & i = 3 \\ (22.5^\circ, 67.5^\circ), & i = 4 \\ (67.5^\circ, 90^\circ), & i = 5 \end{cases} \quad (5)$$

In the above equation, $j \in [1, n]$, $[\cdot]_j$ represents the j^{th} instance of the uniform distribution, and ‘ i ’ represents the standard view. Different random projections of the generated 3D volume S_{ij} are then obtained by feeding these θ_{ij} through the proposed 2D view generator.

$$S_{ij} = \rho(V_{\theta_{ij}}) \quad (6)$$

Finally, to measure the accuracy of the generated 3D silhouette, we compare the generated random projections S_{ij} with the 2D silhouettes I_i corresponding to each of input thermal view T_i . Dice loss, as represented in Eq. (7), that originates from Sorensen–Dice coefficient [29] was used to compare the two image silhouettes.

$$D(I, S) = 1 - \frac{2|I \cdot S|}{|I|^2 + |S|^2} \quad (7)$$

S and I represent the 2D rendered projection of generated 3D volume and 2D silhouette corresponding to the input thermal image, respectively.

2.2.5 Overlay

After achieving loss convergence and obtaining the 3D breast silhouette, overlaying the temperature values is performed for 3D temperature reconstruction. To ensure the preservation of actual temperature values, the overlay of input temperature values onto the generated 3D silhouette is performed as an additional step instead of directly learning the 3D temperature surfaces with 3D-BreastNet. For each input 2D thermal image, the estimated view angle is obtained by identifying the view angle that corresponds to the minimum Dice loss between its 2D silhouette image and different projections of the generated 3D breast silhouette. This estimated view angle compensates for deviations from the standard view angles in the captured images. The generated 3D silhouette is then rotated to the estimated view angle, and the corresponding 2D temperatures from the input images are added to the 3D silhouettes, resulting in the generation of 3D breast surface temperatures. In cases where multiple temperatures are added to the same point, the average temperature is considered. Figure 1 illustrates the interactions between the different modules involved in generating the 3D temperature overlay image.

3 Experimentation and Results

The dataset consisted of thermal data collected from 2027 participants, which was randomly divided into training, validation, and independent test sets with 1627, 200, and 200 participants, respectively. The random seed used for the split was kept consistent

throughout the experimentation process. Each participant's thermal data comprised five captured thermal images corresponding to the standard views. These thermal images were represented as arrays of raw temperature values with dimensions of 320×240 .

To normalize the temperature values within a range of $[0, 1]$, each participant's thermal images were first normalized using their corresponding minimum and maximum temperatures. Subsequently, foreground segmentation was applied to the normalized images to obtain the 2D silhouette images. These images from different views were concatenated to form the input for the 3D silhouette generator architecture. All images were resized to 64×64 to match the input dimensions of the 3D silhouette generator.

The implementation of the 3D silhouette generator (G) employed an encoder-decoder architecture. The encoder consisted of five blocks of convolutional and batch-normalization layers, while the decoder included five blocks of 3D transposed convolutional and 3D batch-normalization layers. The encoder performed down-sampling to map the stacked thermal images to a flattened latent code, which was a 1000-dimensional vector. The decoder, operating in tandem with the encoder, mapped the latent code to the 3D reconstruction model. The final output of the decoder was a 3D volume with dimensions $64 \times 64 \times 64$.

The Dice loss function described in Eq. (7) was used to penalize the 3D silhouette generator (G) when the random 2D projections of the predicted 3D silhouette, obtained from a uniform distribution, deviated significantly from the input 2D thermal images. For experimentation, we set the number of random renderings per view (n) to 2 and the batch size (B) to 16 subjects. The Adam optimizer with a learning rate of 0.025 and decay rates of 0.5 and 0.9 was used for loss optimization. The network was trained for 400 epochs, allowing the network to learn from a diverse range of random projections of the generated 3D silhouette.

Figure 2 depicts the average Dice loss of the validation and training data over the 400 epochs, demonstrating the convergence of the Dice loss and affirming the learning capability of the proposed 3D-BreastNet.

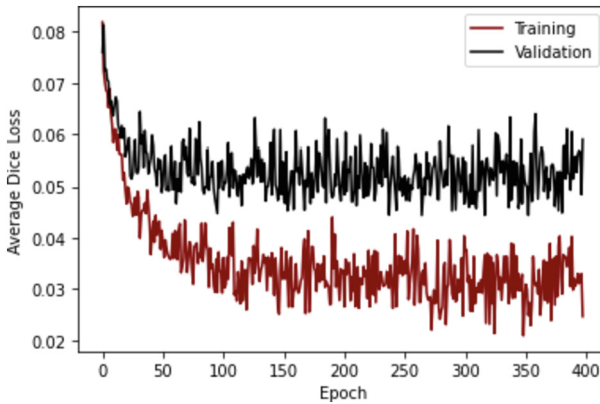


Fig. 2. Illustration of average training and validation loss curves. The training and validation Dice loss/ (1-Dice index) are very close and differ by 2% with validation Dice index as high as 95% (0.95).

Table 1 presents different metrics, such as Dice Index, Jaccard Index, and Hausdorff distance, averaged over the independent test subjects for rendered slices at standard frontal (0°), lateral ($\pm 90^\circ$), and oblique ($\pm 45^\circ$) views. These metrics were calculated assuming that the input images were captured at precise view angles, despite the challenges mentioned in Sect. 2 regarding the imaging protocol. Therefore, the actual results may slightly differ from the reported outcomes. Figure 3 illustrates the variation of Dice loss with slices of the volumes taken at different view angles in comparison to the input images. Furthermore, Fig. 4 showcases the results of the proposed approach for different participants.

Table 1. Evaluation metrics: The mean Accuracy, Dice index, Jaccard index and Hausdorff distance of the 2D projections generated from 3D thermal silhouette at standard view angles (0° , $\pm 45^\circ$ and $\pm 90^\circ$) when compared with the input captured 2D silhouette on the independent test dataset.

	Accuracy	Dice Index	Jaccard Index	Hausdorff Distance
Frontal	0.9838	0.9889	0.9781	1.4142
Oblique	0.9583	0.9690	0.9400	4.5615
Lateral	0.9566	0.9484	0.9022	5.6180
Overall	0.9663	0.9688	0.9401	3.8646

4 Discussion

Breast thermography offers several advantages, including low cost, non-invasiveness, and portability. Recent research [1–3, 9–19] has demonstrated its potential as an alternative imaging modality, particularly when standard imaging techniques like mammography are unavailable. Three-dimensional thermal imaging plays a crucial role in breast diagnosis [3, 16], 3D bio-heat simulation [15], understanding tumor properties [32, 33], and breast augmentation [19], enabling better visualization. By reconstructing 3D temperature information from 2D thermal images alone, additional hardware for capturing depth information is not required. However, generating 3D breast thermal images presents various challenges. Unlike visual images, thermal images lack strong scale-invariant features, and obtaining precise 3D ground truth breast surfaces for supervised learning is difficult. Moreover, imperfect image capture and lack of information on the exact view angle further complicate the problem.

Our proposed solution, 3D-BreastNet, leverages the strengths of deep learning to address these challenges. It generates the 3D breast thermal surface from 2D thermal images without requiring explicit view angles or actual 3D ground truth. To assess the accuracy of 3D-BreastNet, we compared random 2D projections of the 3D silhouette at standard angles with the input images. While, in a real scenario, comparing predicted 2D projections at each angle with actual 2D projections would be ideal, we considered comparisons at the five standard view angles at which the images were captured. This

approach provides a holistic view of performance, as these angles are widely spaced, making it difficult for the 3D generator to overfit to these specific view angles without disregarding rotational geometry constraints.

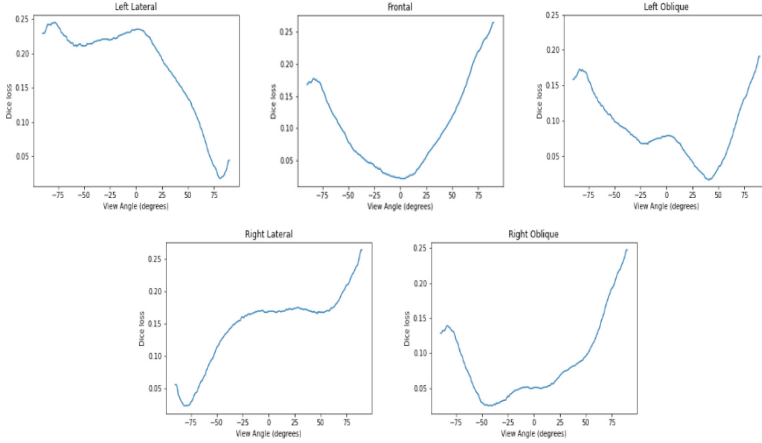


Fig. 3. Variation of averaged Dice loss for the input 2D silhouette images at a) frontal (0°), b) left lateral (90°), c) left oblique (45°), d) right lateral (-90°), and e) right oblique (-45°) views when compared with 2D random projections of their 3D generated volumes at view angles varying from -90° to $+90^\circ$ at 1° intervals on the independent test set.

Our main hypothesis in reconstructing 3D breast thermal silhouettes from 2D thermal images is that the Dice loss between the 2D projections of the 3D breast silhouette and the input 2D silhouette increases when the projection angle deviates significantly from the captured view angle. With this assumption, we trained G for longer epochs to minimize the Dice loss between the input image and its neighbouring random views, aiming for the loss to be optimal when the expected predicted random slices of the volume closely match the actual input image for that view angle. To validate the hypothesis, we calculated the expected (mean) Dice loss for each standard view by comparing the 2D silhouettes of thermal images in the independent test set with different projections of the generated 3D silhouette volumes at view angles ranging from -90° to $+90^\circ$ at every 1° interval. The graphs in Fig. 3 illustrate that the average Dice loss was at its minimum when the rendered 2D projections were close to the standard view angles (0° , $\pm 45^\circ$, and $\pm 90^\circ$). By examining Fig. 3, we determined that the average Dice loss was lowest at 4° , -80° , $+81^\circ$, -44° , and $+41^\circ$ on the test dataset for frontal (0°), right lateral (-90°), left lateral ($+90^\circ$), right oblique (-45°), and left oblique ($+45^\circ$) views, respectively. These angles slightly deviated from the standard view angles, and we believe they represent the mean view angles at which the input images were captured, accounting for the challenges faced during image capture.

The average Dice loss of 5% on the test and validation datasets demonstrates the ability of 3D-BreastNet to generate 3D structural volumes that closely resemble true 3D volumes. Additionally, as shown in Fig. 4(c), the rendered 2D projections of the 3D breast

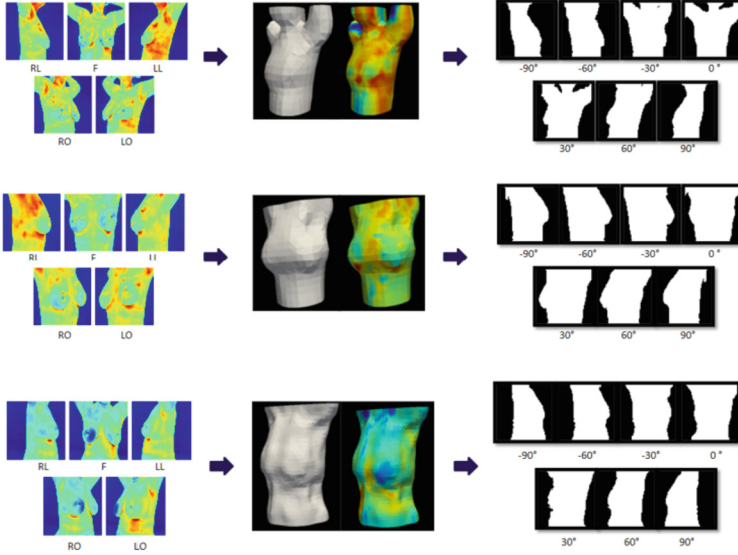


Fig. 4. 3D-BreastNet results for independent test subjects: a) Captured input thermal images, b) Generated 3D silhouettes and their corresponding overlaid 3D temperature surfaces c) 2D projections of the generated 3D silhouette in (b) at view angles -90° , -60° , -30° , 0° (top row) and 30° , 60° , 90° (bottom row).

silhouette on the test set at 30-degree intervals accurately reflect the rotational movements and visually correlate with the 2D silhouettes of the input thermal images. For generating 3D temperature images, we used a simple overlay algorithm that minimally alters the true temperatures. As the 3D-generated silhouettes closely resemble actual structures, the 3D temperature images are expected to approximate true 3D temperatures due to this overlay. Notably, our approach does not involve registration between multiple hardware devices, simplifying clinical integration compared to existing methods that utilize 3D scanners [20–24].

Overall, the proposed methodology of 3D-BreastNet yields reasonably good results, but it has a few limitations. One significant limitation is the lack of restrictions on the breast shapes of the generated volume. As a result, the predicted structures may appear distorted in some cases, as shown in Fig. 4(b). The breast surface can be assumed to closely resemble a semi-ellipsoidal structure. In our future work, we plan to incorporate this shape prior into the generator network to reduce deformations in the predicted volumes. We also intend to explore the adoption of concatenating encoder outputs with decoder inputs, as proposed in U-Net [34], to improve the generation of 3D breast silhouettes.

Three-dimensional thermal imaging is equally important for imaging other body parts, such as the hand, face, and knee [20, 35]. The challenges of obtaining 3D labeled ground truth, requiring additional hardware, and lacking precise view angle information are common in the 3D reconstruction of these body parts as well. Since our proposed 3D-BreastNet can alleviate these issues, we believe the algorithm can be extended to

enable 3D thermal reconstruction of these body parts. In our future work, we aim to validate the performance of our approach for 3D thermal reconstruction of these body parts.

5 Conclusion

In conclusion, thermography for breast imaging is gaining renewed attention due to its numerous advantages, including affordability, non-invasiveness, and portability. However, it's important to note that thermal images only provide a two-dimensional view of the breast surface temperature. To overcome this limitation, our 3D-BreastNet aims to generate three-dimensional (3D) surface temperatures from two-dimensional (2D) thermal images. Notably, this is the first approach that achieves 3D breast surface image generation from 2D breast thermograms without requiring additional imaging hardware or precise view angle information. The proposed self-supervised loss function is innovative and holds potential for generating 3D surfaces with other imaging modalities. Future work should focus on incorporating shape priors into the deep learning model, which will further enhance its capabilities. Overall, our research presents a promising direction for improving breast imaging through the generation of 3D surface temperatures, opening doors for more accurate and comprehensive analysis of breast health.

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